

Network Security Concepts — Guided Notes ANSWER KEY

Unit 4.1.1 | CompTIA Network+ N10-009 Obj. 4.1 | N10-008 4.1

For instructor use / distribute after students complete the notes.

Question 1

The three pillars of the CIA Triad are _____, _____, and _____. Briefly define each in one sentence.

Answer: Confidentiality (only authorized parties can read data), Integrity (data has not been altered in transit or at rest), Availability (systems and data are accessible when needed).

Question 2

A _____ is someone with authorized network access who misuses it, while a _____ is an outside attacker with no prior access. Which poses the greater risk in most enterprise environments, and why?

Answer: Insider threat; external threat. Insiders often pose greater risk because they already have valid credentials, know the network layout, and their activity blends with normal traffic.

Question 3

Explain the difference between a vulnerability, a threat, and a risk. Give a one-sentence example of each in a network context.

Answer: Vulnerability: a weakness (e.g., unpatched switch firmware). Threat: something that can exploit it (e.g., an attacker scanning for that CVE). Risk: the probability and impact if exploited (e.g., full switch compromise exposing all VLANs).

Question 4

Zero-day vulnerabilities are dangerous because they have _____ available from the vendor when attackers first exploit them. The primary defense while a patch is unavailable is called _____.

Answer: No patches; compensating controls (e.g., firewall rules blocking the vulnerable service, disabling the feature, network segmentation).

Question 5

Defense in depth means using _____ layers of security controls so that failure of any single control does not result in a complete breach. Name three different layers this strategy might include.

Answer: Multiple; examples include: perimeter firewall, internal segmentation, IDS/IPS, endpoint antivirus, MFA, physical locks, and user training.

Question 6

The principle of least privilege states that users and systems should have access to only what they _____ to perform their job. How does this limit the damage from a compromised account?

Answer: Need (minimum necessary). A compromised account with limited privileges can only access a small subset of resources, reducing the blast radius of the breach.

Question 7

Network segmentation limits _____ movement by placing critical systems in separate VLANs or security zones. What firewall behavior enforces the boundary between zones?

Answer: Lateral; inter-VLAN firewall rules (or an internal segmentation firewall) that explicitly permit only required traffic flows and deny everything else.

Question 8 — Real World Application

REAL WORLD: A school's student VLAN can reach the staff grading portal because no ACL separates them. Which CIA Triad principle is most directly violated, and what single technical control would fix it?

Answer: Confidentiality — student grades and staff data should not be readable by unauthorized users.

Fix: an inter-VLAN ACL (or firewall rule) explicitly blocking student VLAN from reaching the staff server subnet.